CSE

Fundamentals of AIML

AI Focus & Habit Intelligence System

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Introduction

In today's digital age, students face major challenges in maintaining focus and discipline due to distractions such as excessive phone usage, poor sleep habits, and lack of structured routines.

Traditional productivity tools focus only on scheduling tasks but fail to measure **actual cognitive focus and behavioral discipline**.

The **AI Focus & Habit Intelligence System** is designed to solve this problem by combining:

- Cognitive performance testing (memory-based game)
- Habit analysis (daily lifestyle inputs)
- AI-based evaluation (focus + discipline scoring)

This system provides **real-time feedback, performance analysis, and improvement suggestions**, helping users build better habits and improve concentration.

Problem Statement

Students struggle with:

- Maintaining consistent focus during study sessions
- Understanding how daily habits affect productivity
- Lack of self-awareness about discipline levels
- No simple tool that combines **focus measurement + habit analysis**.

Problem:

How can we design a system that:

- Measures a user's **focus level**
- Analyses their **daily habits**
- Provides **intelligent feedback and suggestions**
- Helps improve productivity in a simple and interactive way?

Why This Problem Matters:

In today's digital environment, students are constantly exposed to distractions that reduce their ability to focus. Understanding the relationship between daily habits and cognitive performance is essential for improving productivity. This project aims to bridge that gap by combining behavioral analysis with real-time focus measurement.

Objectives

- ❖ To develop a GUI-based system using Tkinter .
- ❖ To measure focus using a memory-based game .
- ❖ To collect habit-related inputs from users .
- ❖ To calculate:
    - Focus Score
    - Discipline Score
- ❖ To generate AI-based feedback .
- ❖ To visualize performance using graphs.
- ❖ To provide personalized improvement suggestions.

Functional Requirements

1. Major Modules

❖ **Start Module**

- Displays welcome screen .

❖ **Game Module**

- Shows number/color sequences
- User recalls sequence
- Score calculated based on accuracy .

❖ **Difficulty Module**

- Easy / Medium / Hard levels .

❖ **Habit Question Module**

- Study hours
- Sleep hours
- Phone usage
- Exercise

❖ **Analysis Module**

- Calculates focus & discipline scores

❖ **Result Module**

- Displays:
  - Scores
  - Feedback
  - Suggestions

❖ **Graph Module**

- Shows bar graph using matplotlib .

2. Inputs

- User memory response (game input)
- Study hours
- Sleep hours
- Phone usage
- Exercise duration

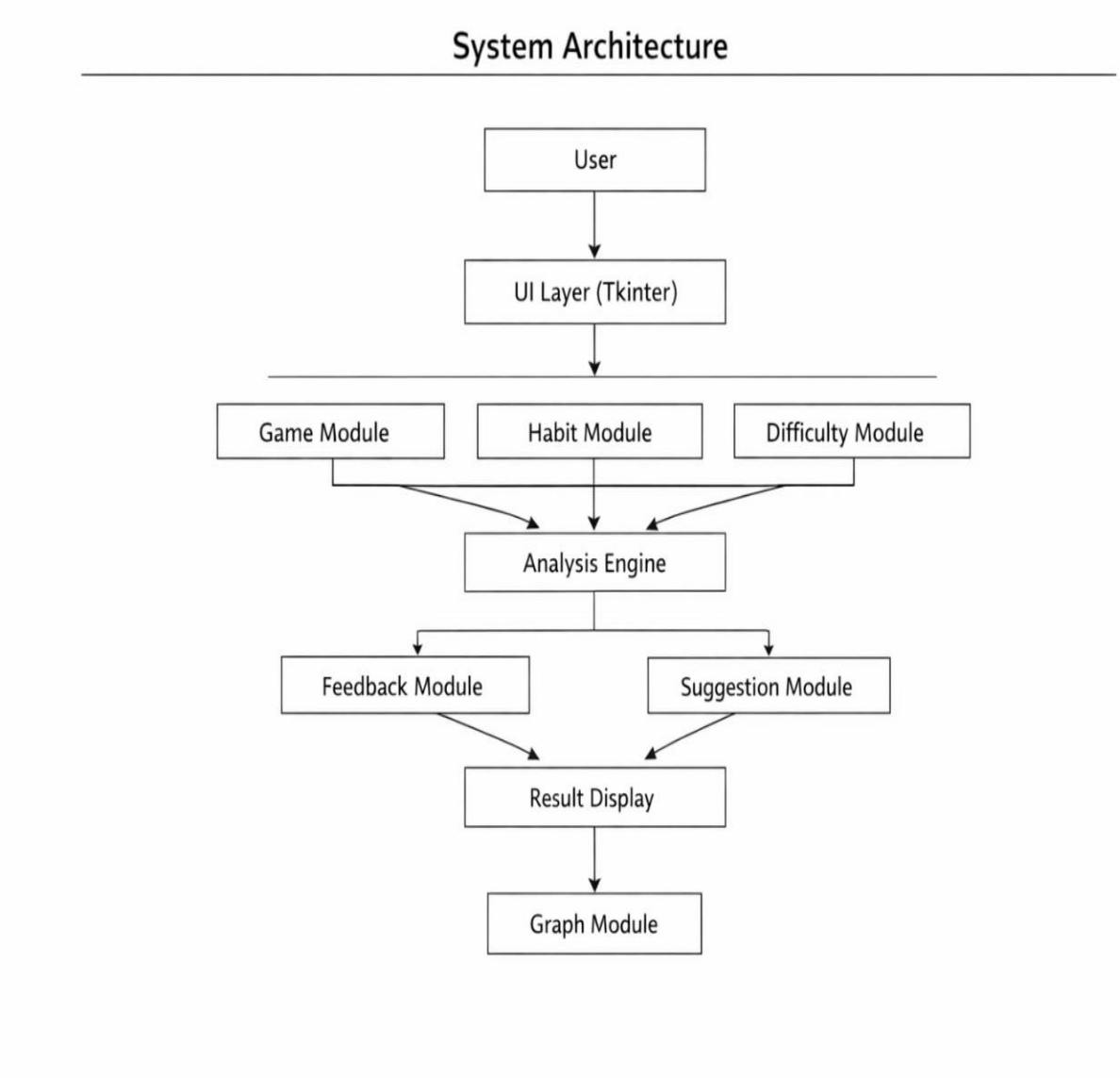
3. Outputs

- Focus Score
- Discipline Score
- AI Feedback (Good / Improve)
- Suggestions
- Performance Graph

Non-Functional Requirements

- **Usability:** Simple and clean GUI
- **Performance:** Fast execution
- **Reliability:** Accurate scoring logic
- **Maintainability:** Modular functions
- **Scalability:** Can add ML/AI features later
- **Aesthetics:** Structured UI with smooth navigation

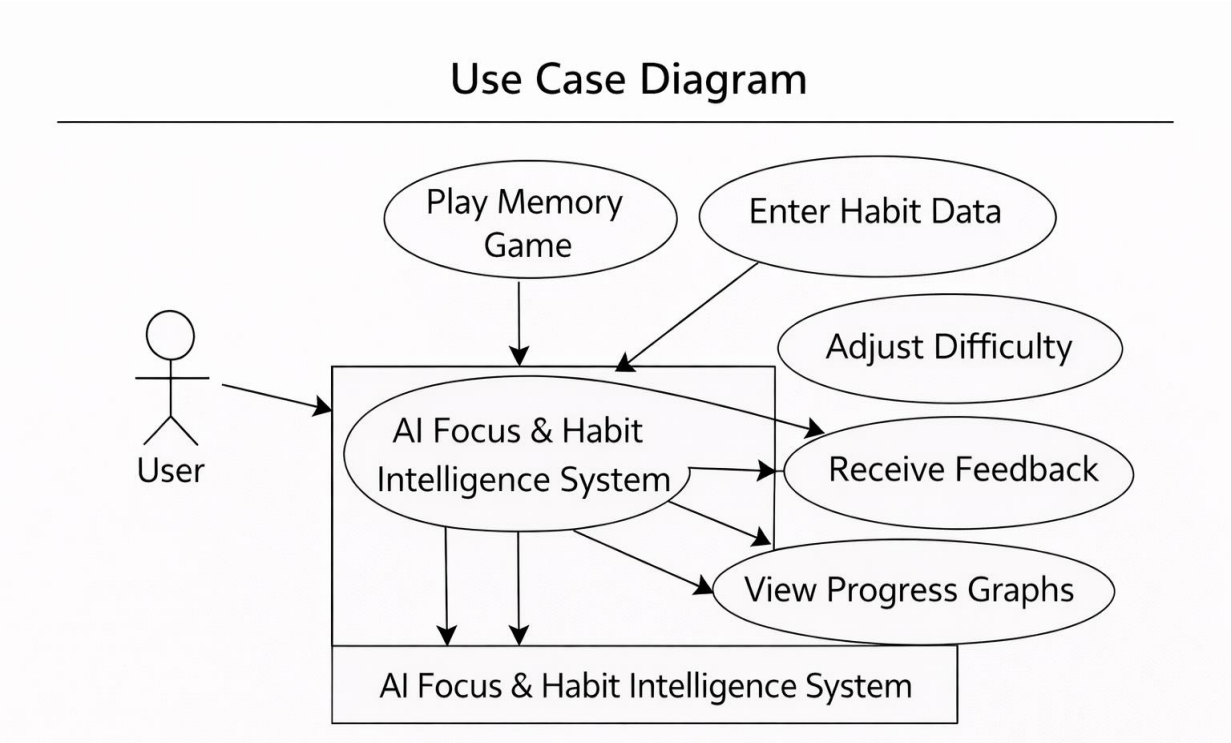
System Architecture



This diagram represents the overall structure of the system, showing how different modules such as the user interface, game module, habit module, and analysis engine interact with each other. The architecture illustrates the flow of data from user input to processing, feedback generation, and graphical output.

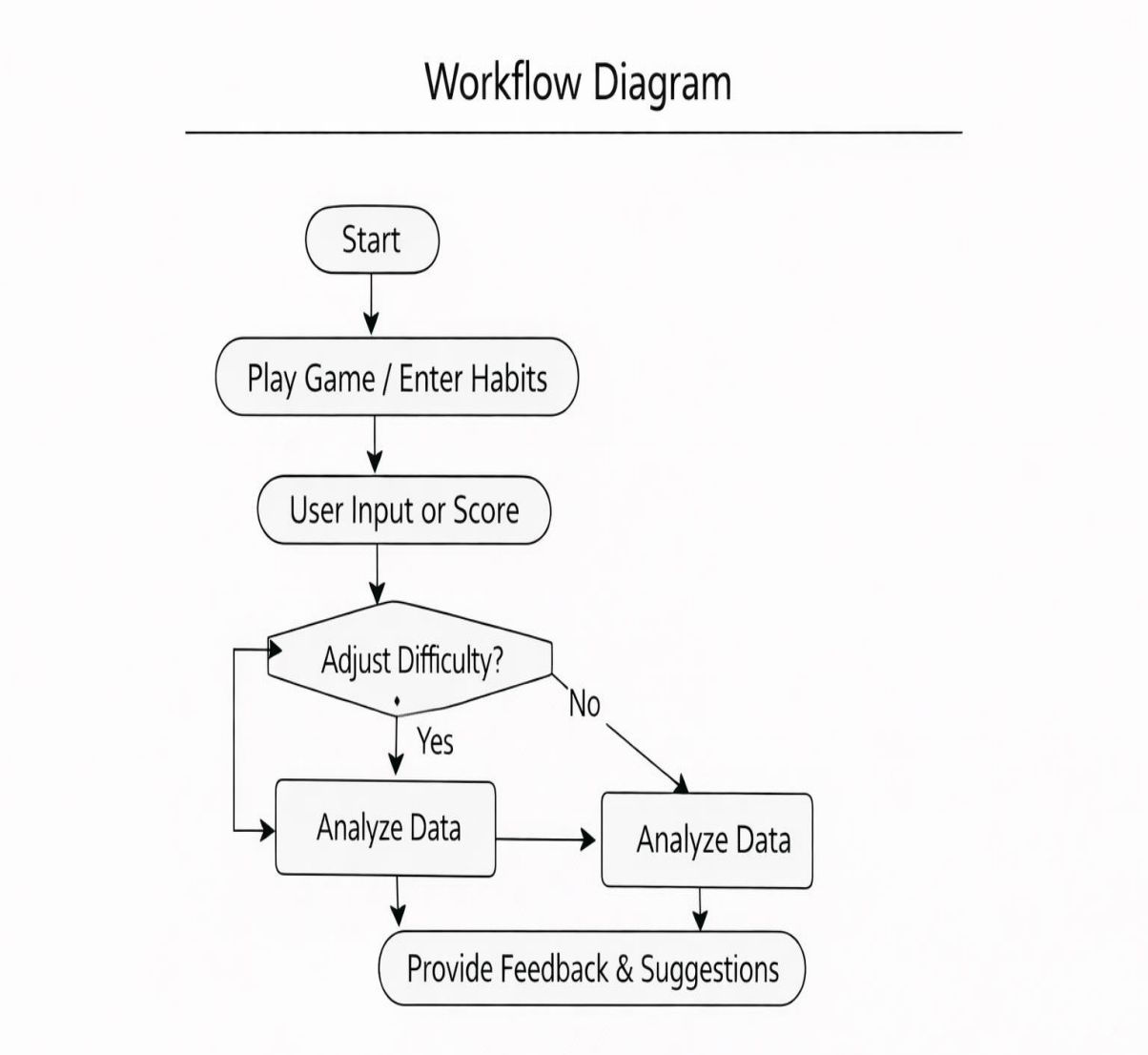
Design Diagrams

1. Use Case Diagram



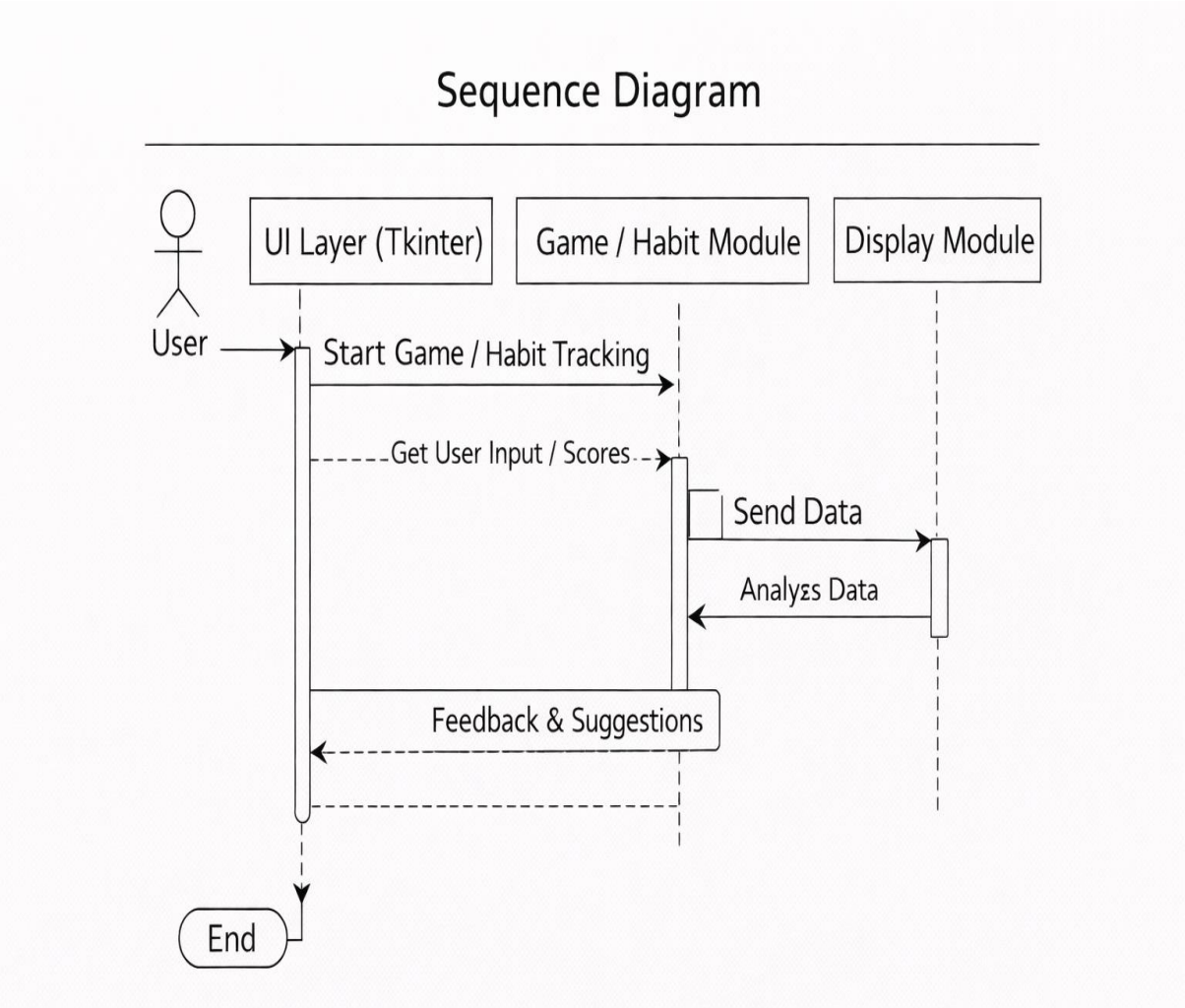
The use case diagram shows the interaction between the user and the system. It highlights the main functionalities, including playing the memory game, entering habit data, adjusting difficulty, receiving feedback, and viewing progress graphs.

2. Workflow Diagram



This diagram illustrates the step-by-step process of the system, starting from user input to final result generation. It includes key stages such as game execution, habit input, data analysis, and feedback generation.

3. Sequence Diagram



The sequence diagram depicts the interaction between different system components over time. It shows how the user, UI layer, modules, and analysis engine communicate to process inputs and generate feedback and results.

4. Class/Structure

Main Components:

- GUI Frames
- Game Logic
- Input Handler
- Analysis Engine
- Graph Module

Design Decisions & Rationale

- ❖ Tkinter used for simplicity and fast GUI development
- ❖ Memory game chosen to simulate **real cognitive focus**
- ❖ Difficulty levels added for adaptability
- ❖ Score formula balances:
    - Game performance (focus)
    - Lifestyle habits (discipline)
- ❖ Graph included for visual understanding
- ❖ Multi-screen UI improves user experience

Implementation Details

Technologies Used:

- Python 3.x
- Tkinter
- Matplotlib
- Random Modul

Core Logic:

- **Focus Score Calculation-**
  focus = (game_score * 0.7) + (discipline * 0.3)

- **Discipline Score Logic-**

- Study ≥ 6 → High
- Sleep ≥ 7 → Good
- Phone usage low → Better discipline
- Exercise → Positive impact

AI-Based Decision Logic:

Although the system does not use advanced machine learning algorithms, it incorporates rule-based intelligent decision-making. The system analyzes user performance in the memory game along with habit-related inputs such as study hours, sleep, and phone usage. Based on predefined conditions, it generates personalized feedback and suggestions. This behavior mimics basic artificial intelligence by adapting responses according to user data and performance patterns.

Results

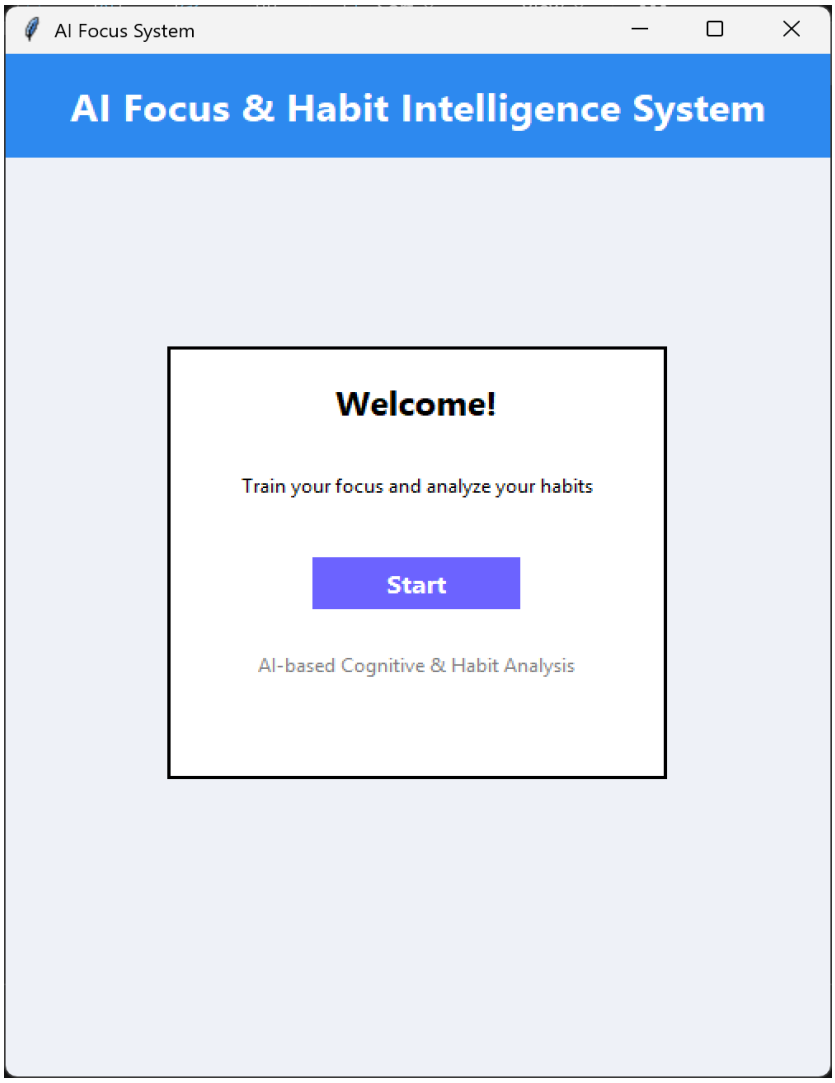
My system produces:

- ❖ Interactive GUI screens:
  - Start screen
  - Game screen
  - Question screen
  - Loading screen
  - Result screen
- ❖ Final output includes:
  - Focus Score
  - Discipline Score
  - Feedback
  - Suggestions
  - Graph visualization

Screenshots of Output -

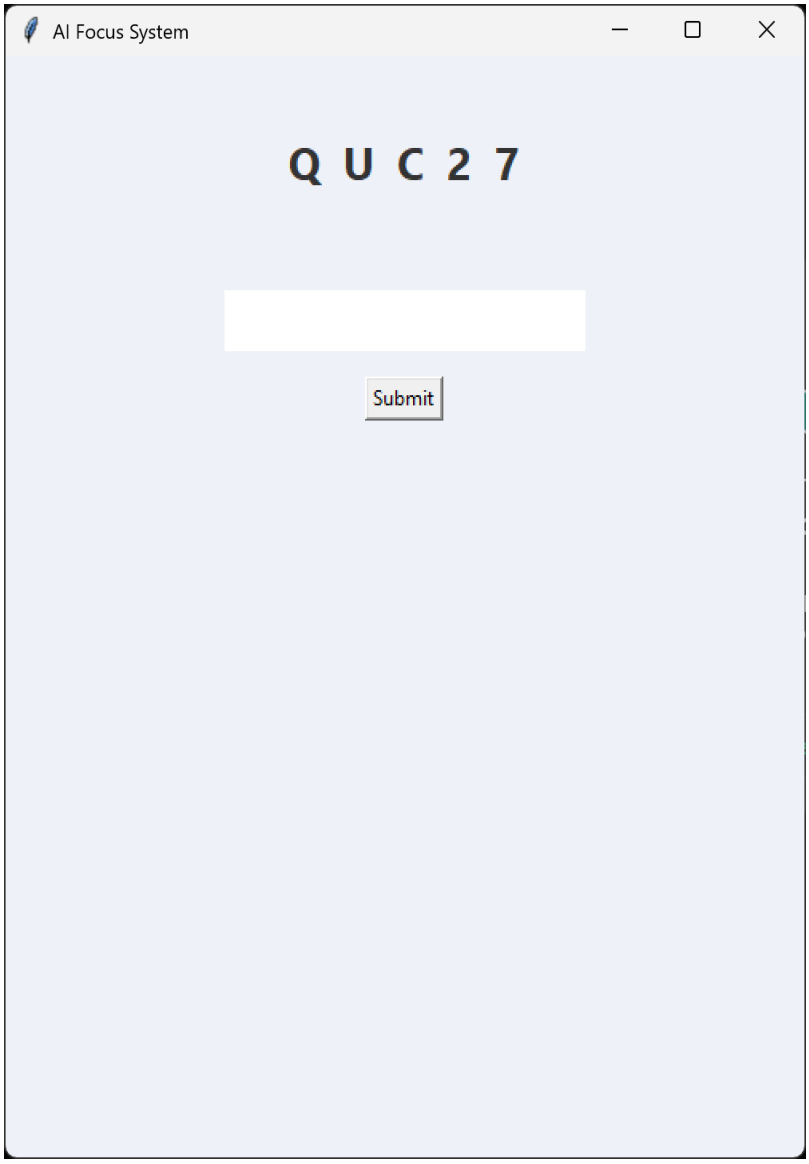
This section presents the actual output of the system through various screenshots, demonstrating the working and functionality of the application.

Start Screen of Application :

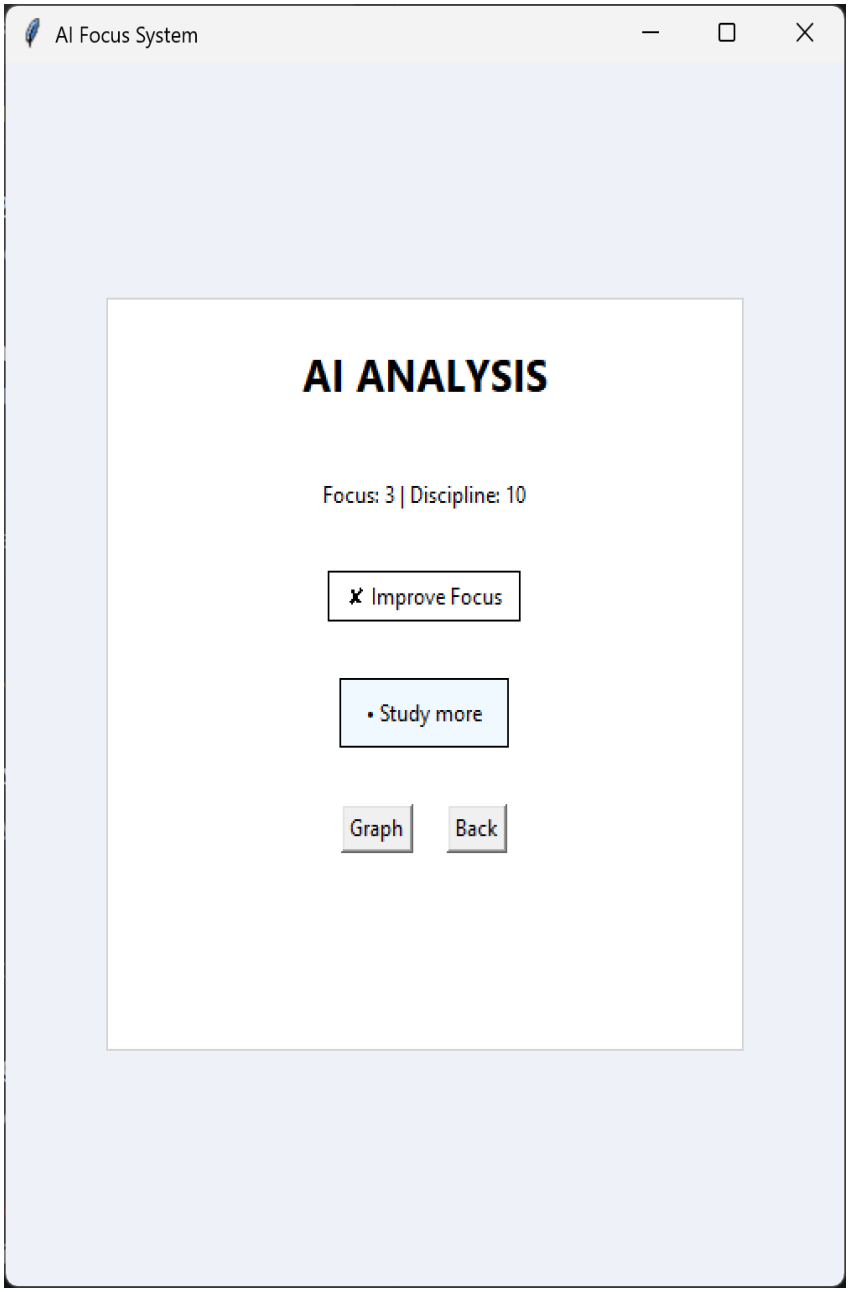


Difficulty Selection Screen:

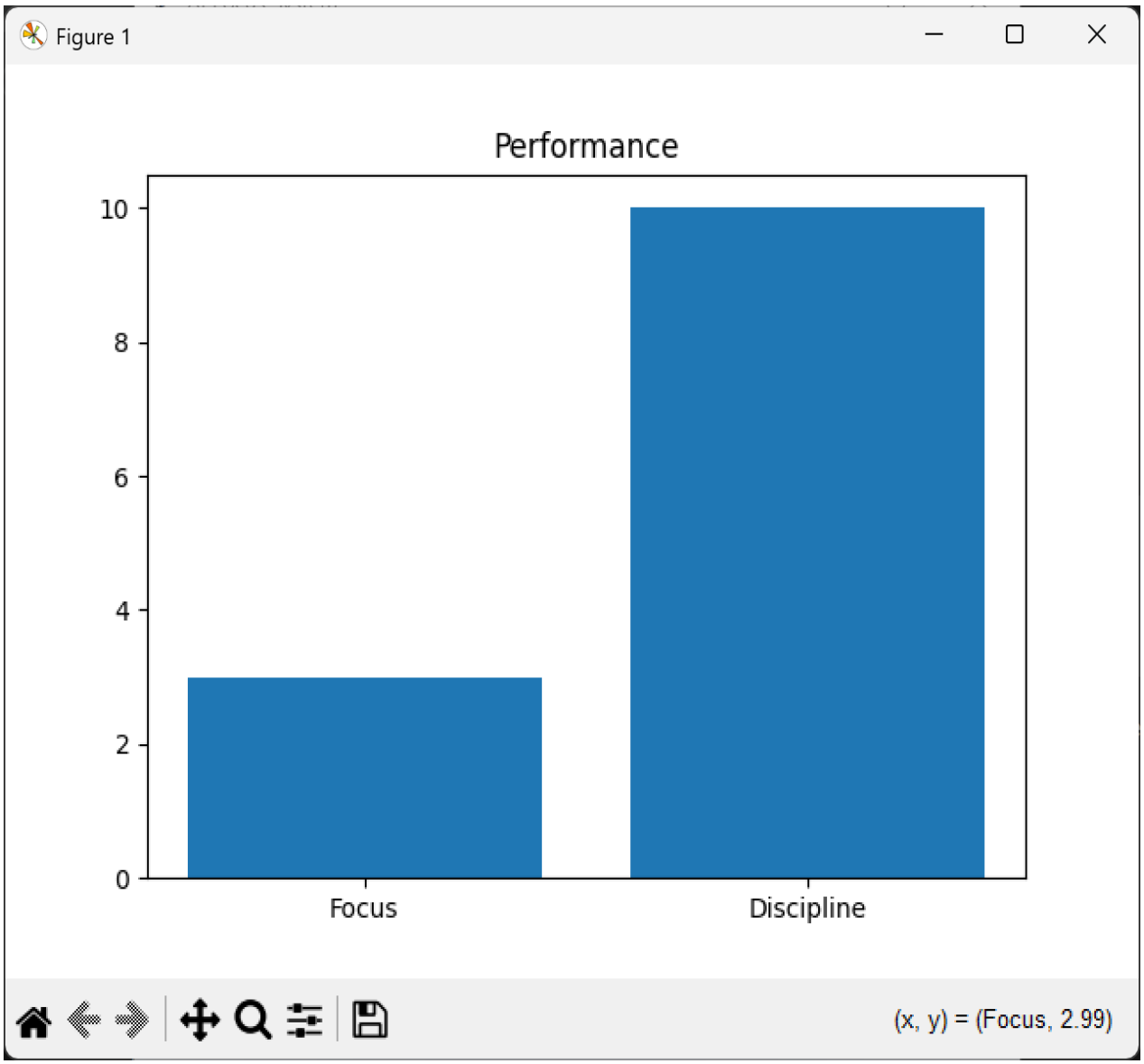
Memory Game Interface:



Result Display Screen:



Performance Graph Visualization:



Testing Approach

Test Case	Input	Expected Output	Result
Valid game input	Correct sequence	High score	PASS
Wrong input	Incorrect sequence	Low score	PASS
High study hours	7+	Good discipline	PASS
High phone usage	>8 hrs	Low discipline	PASS
No exercise	0	Suggestion shown	PASS

Challenges Faced

- Designing multi-screen GUI in Tkinter
- Managing state between frames
- Implementing memory game logic
- Balancing scoring formula
- Handling user inputs properly

Learnings & Key Takeaways

- ❖GUI development using Tkinter
- ❖Event-driven programming
- ❖Logic building and condition handling
- ❖User experience design
- ❖Data visualization using matplotlib
- ❖Real-world problem solving

Future Enhancements

- Machine Learning-based prediction
- User data storage (database)
- Daily progress tracking
- Mobile app version
- Personalized AI coach
- Notifications & reminders

References

- Python Documentation
- Tkinter Docs
- Matplotlib Docs
- StackOverflow
- Course Materials

Conclusion

The project successfully demonstrates how programming concepts can be applied to solve real-life problems.

By combining **cognitive testing + habit analysis**, the system provides a unique approach to improving productivity and self-discipline.

It is simple, effective, and can be expanded into a full AI-based productivity system in the future.